ASSEMBLY GUIDE

Smart Sensor & Display Device

Attention: The print profile is set to pause at layer 180. This allows you to insert two 15-gram weights into the designated slots opposite the dome, directly inside the circular body. Once placed, printing can resume.

This ensures the object floats correctly in the water by balancing the weight of the dome and internal electronics.

1. REQUIRED MATERIALS & COMPONENTS

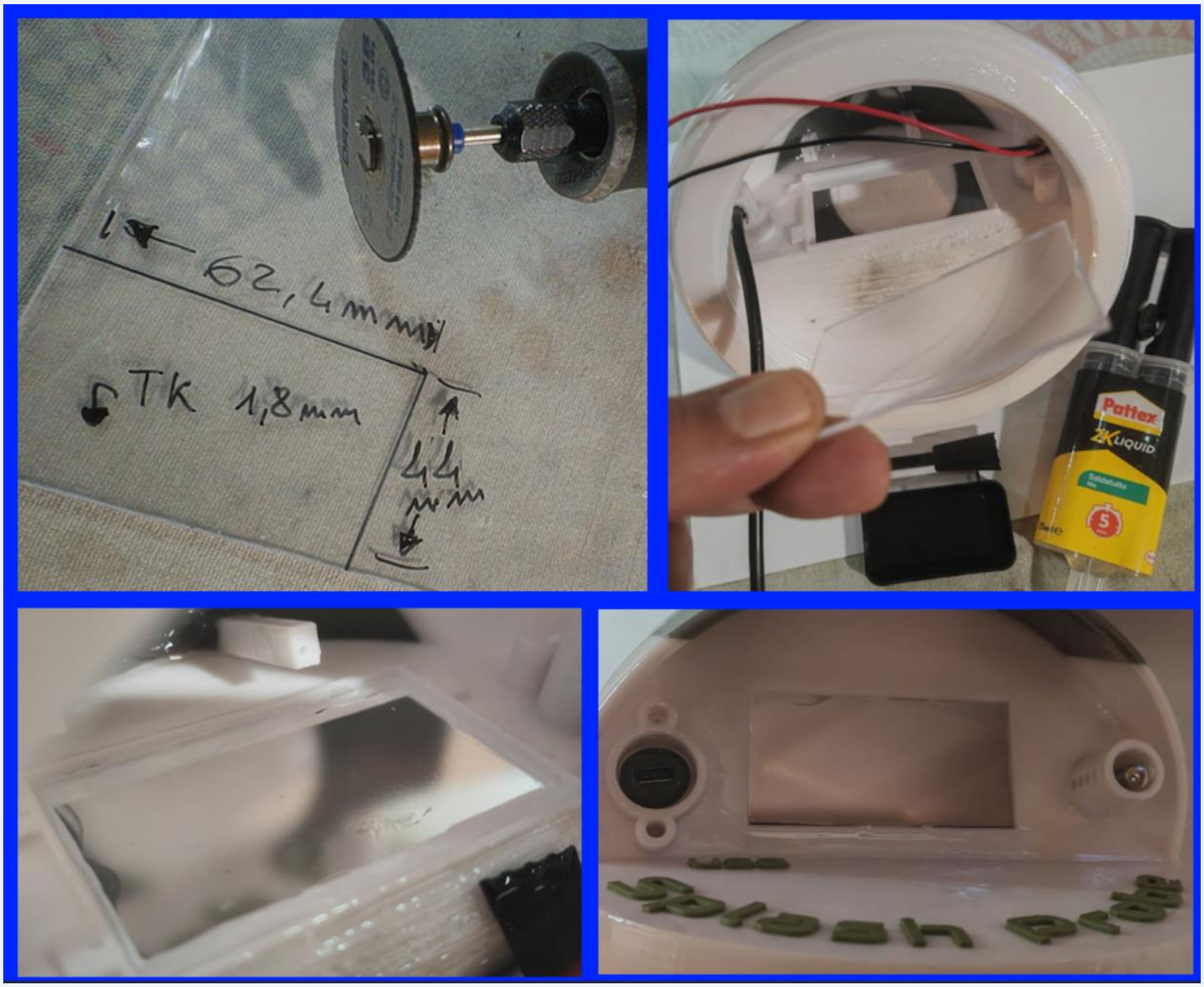
COMPONENT / MATERIAL	DETAILS / SPECIFICATIONS
Two 15g weights each, to be concealed/embedded inside the body during printing.	
3D Printed Enclosure	Upper main dome with mounting slots & lower circular base
Plexiglass Panel	Dimensions: 44 mm x 62.40 mm | Thickness: 2 mm
Polymer Adhesive	Transparent, waterproofing adhesive suitable for Plexiglass & PLA
Two-Component Epoxy Resin	For sealing, waterproofing, and surface finishing + 1 paint brush
Electronic Components	• Heltec CubeCell AB01 v2 microcontroller board • 2.13" ePaper display • 12-LED NeoPixel ring • 2x DS18B20 temperature sensors (Air & Water) • 18650 LiPo battery with JST-SH2 connector • 330 Ω resistor | 4.7 kΩ resistor • Connecting wires & circuity cables • USB Type-C connector.
Hardware & Gaskets	• 10 tiny screws (for display, board, and USB cap) • Large TPU gasket (main thread seal)

• Small TPU gasket & USB port protective cap
• Plastic zip-ties for securing battery and cable routing

2. STEP-BY-STEP ASSEMBLY PROCEDURE

STEP 1: Plexiglass Window Bonding

Glue the Plexiglass panel (44 mm x 62.40 mm) into the dedicated inner frame of the 3D-printed top dome using the transparent polymer adhesive. Ensure it is centered properly and clean of excess glue.



Before fixing the temperature sensors in place, how do you identify them?

The two sensors operate on a single **OneWire** channel. Each sensor has its own unique address, and normally these addresses are used to distinguish between them. It is also possible to assign them during the development/pre-compilation stage.

For anyone who wants to **flash the board without modifying the code**, the firmware assigns the **air temperature measurement to the first sensor detected** and the **water temperature measurement to the second sensor detected**.

In **debug mode**, all information is printed to the serial port. If both sensors are detected correctly, the temperatures will also be displayed on the screen, in addition to the serial output.

Gently warm one of the sensors to see its temperature rise and determine which sensor has been assigned by the system as **Air** and which as **Water**. Once identified, you can proceed with installation: the **air temperature sensor** should be mounted on the front panel, while the **water temperature sensor** is located on the circular bottom section underneath the LiPo battery.

In extreme cases, the sensors can be swapped in the code and the firmware can be compiled again.

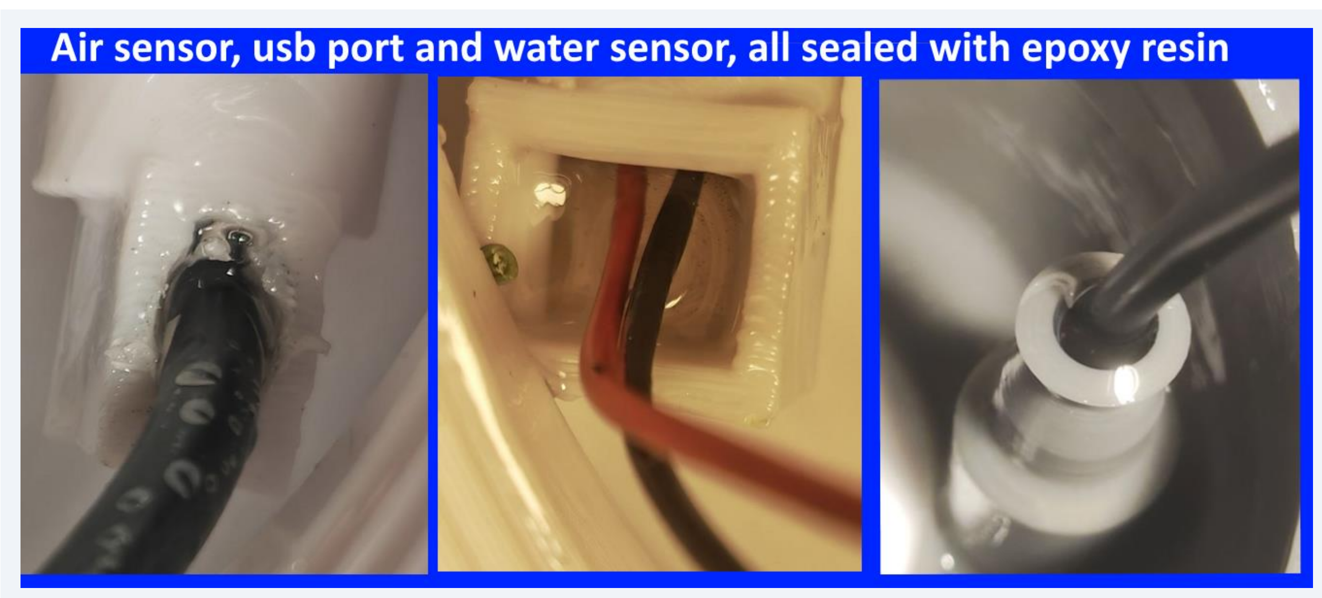
All source code is available online.

STEP 2: USB Port & Air DS18B20 Sensor Installation (Dome)

With the top dome kept upside down, insert the USB port and the first DS18B20 temperature sensor (air sensor) from the outside. Pour two-component epoxy resin into the internal recesses to permanently seal both components.

STEP 3: Second DS18B20 Sensor Installation (Base)

Insert the second DS18B20 sensor into the circular base from below. Seal the cable entry hole completely by pouring epoxy resin into the internal potting cavity within the base.

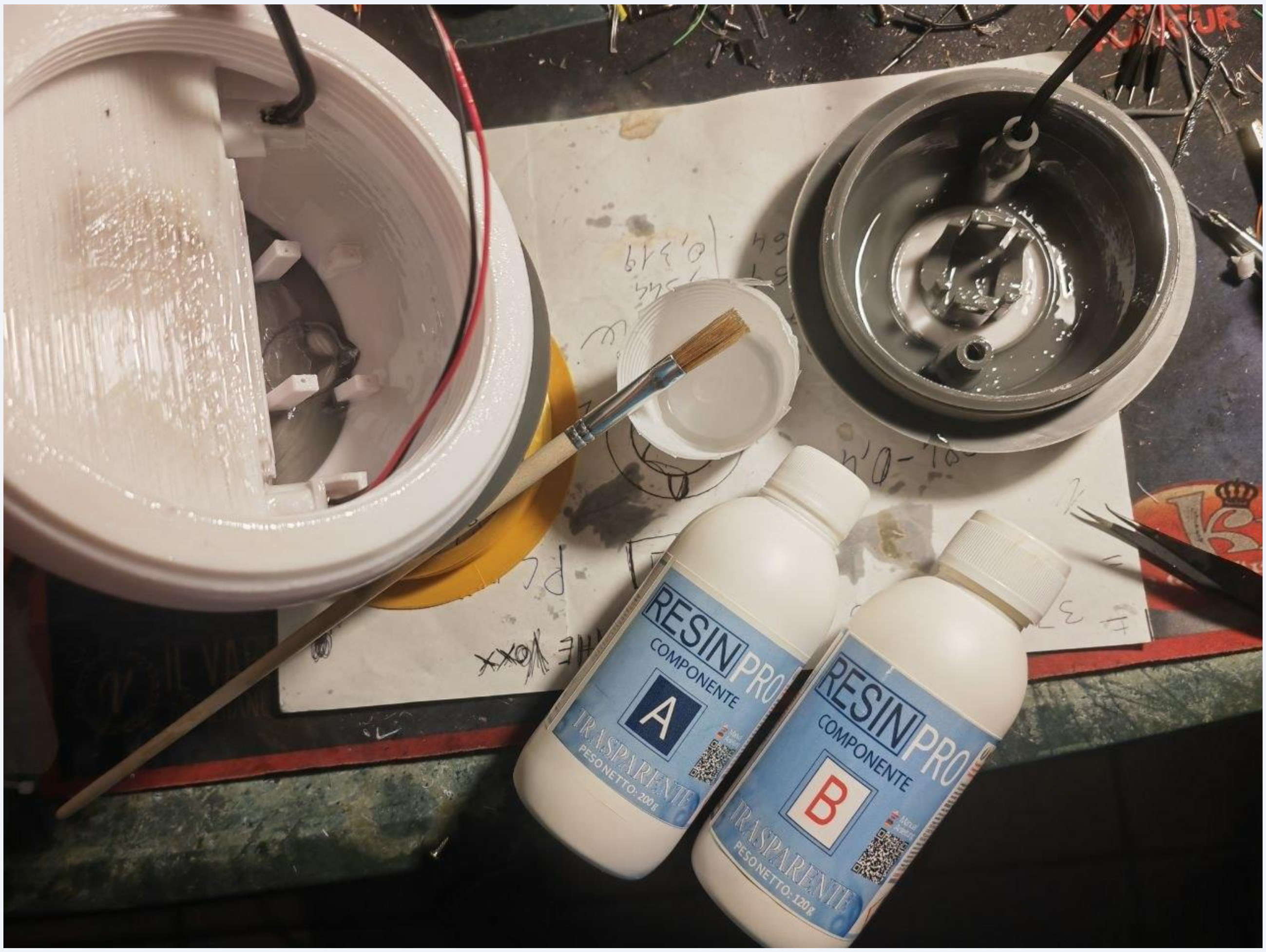


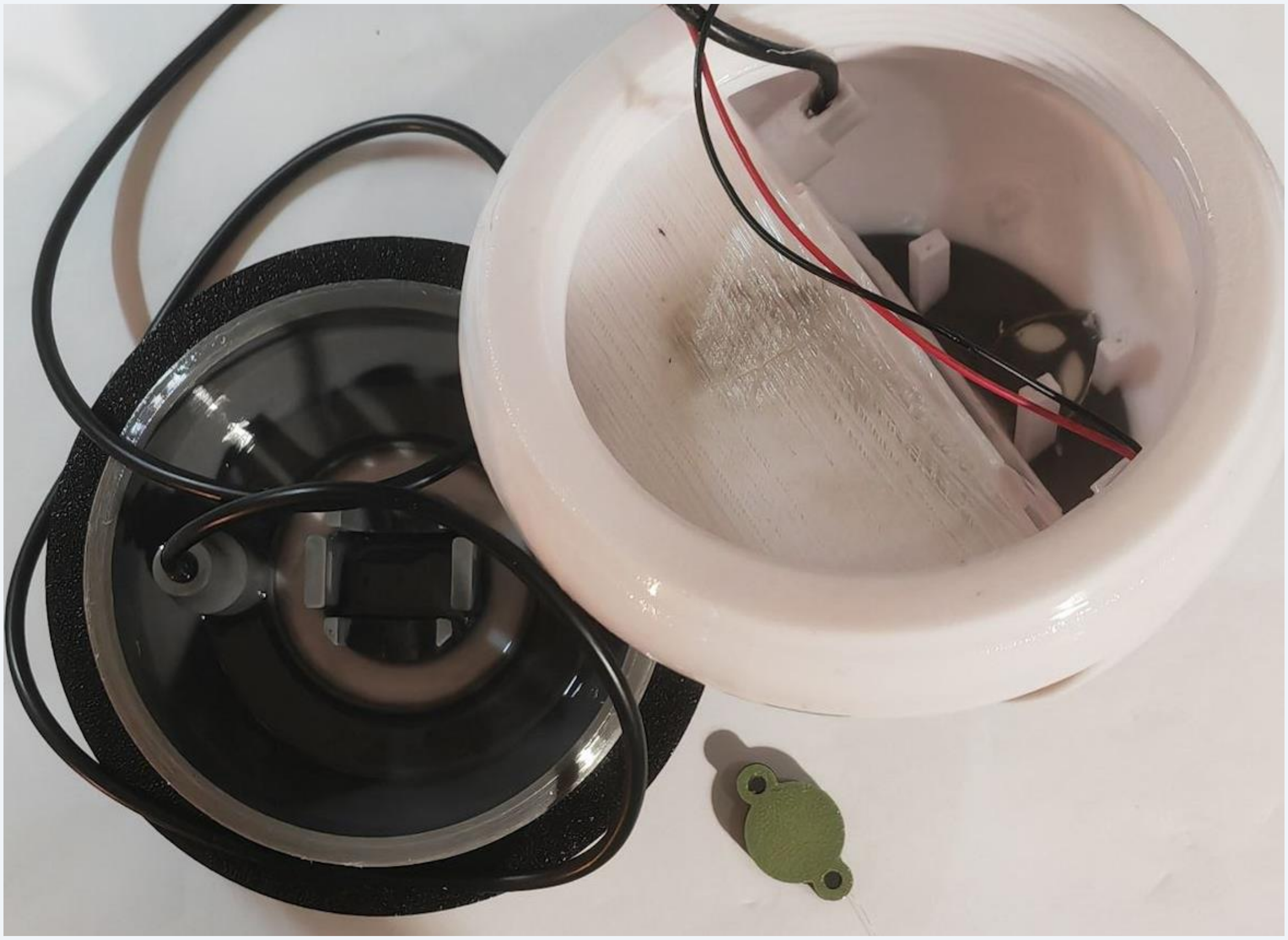
STEP 4: First Curing Phase (24 Hours)

Allow the components to cure in a dry environment for **at least 24 hours** to ensure full hardening of both the epoxy resin and the polymer glue.
Warning: Do not disturb the components during this curing period to avoid breaking airtight seals.

STEP 5: Waterproofing & Epoxy Coating

Using a paintbrush, apply an even coat of epoxy resin over the entire object (both upper dome and lower base, inside and outside).
• Pay close attention to coating the acrylic window joints thoroughly.
• IMPORTANT: Do NOT apply resin to the main center screw threads.
This process makes the object fully WATERPROOF, structurally stronger, and gives it a glossy finish. Apply a second coat after a few hours if necessary.





STEP 6: Second Curing Phase (24 Hours)

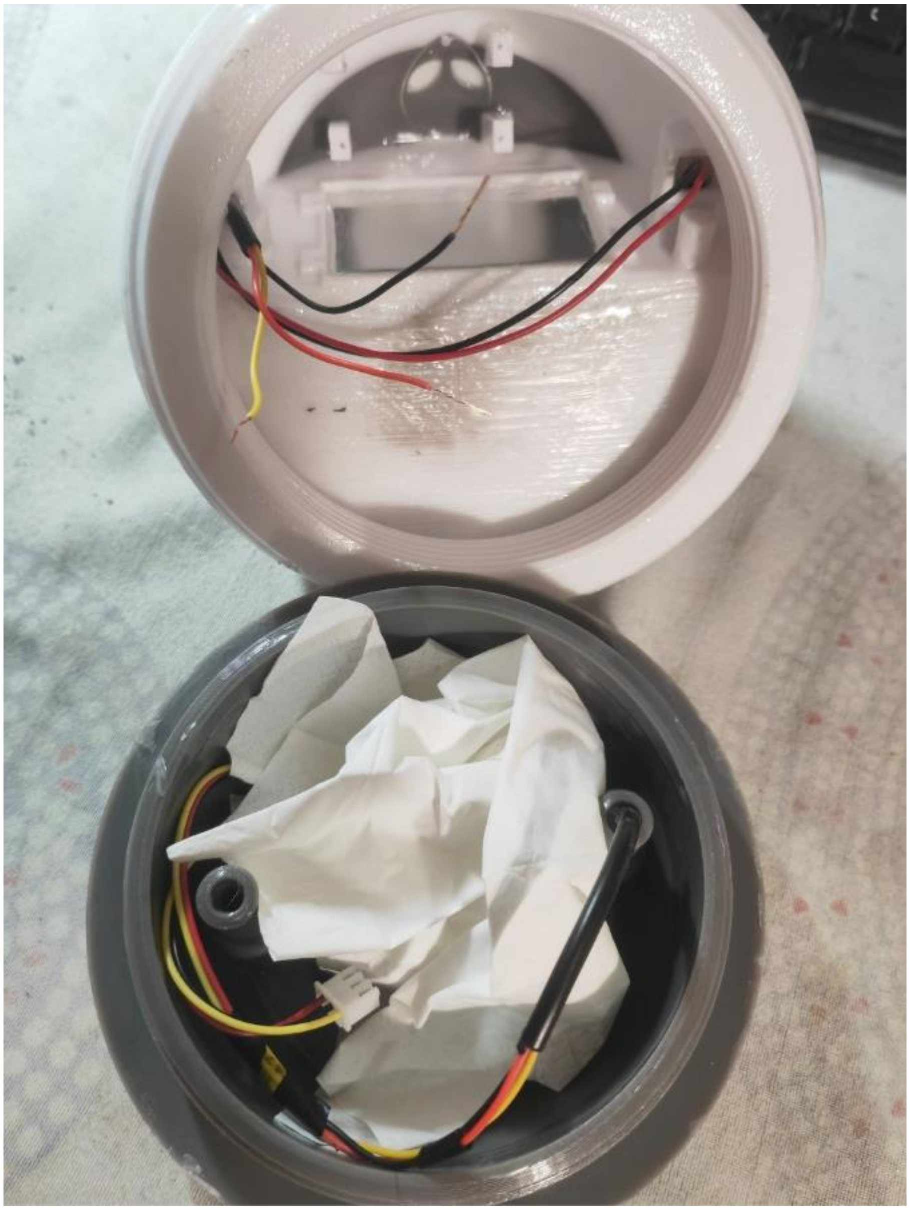
Wait **at least another 24 hours** for the external epoxy coating to dry and cure completely before installing internal electronics.

** WATERPROOFNESS TEST

If everything has been done correctly, the object should be able to withstand even very brief immersion. Since this is a handmade item, it is best to check the watertightness before installing the electronics. Close the USB port with the gasket, cap, and two screws; install the main large gasket between the two bodies and tighten everything firmly until sufficient pressure is applied to the gasket, ensuring a watertight seal.

BASE WATER TEST (without electronics; only the DS18 sensor is installed and sealed with resin):

Complete test. I added a piece of paper towel to detect any absorbed water:




Splash Probe

STEP 7: Firmware Upload

Connect the Heltec CubeCell AB01 v2 microcontroller board to your computer via USB and upload the device firmware.

Complete Project with source code and firmware file at:
https://github.com/Nando75/SplashProbe

Directly Flash Tool:
https://github.com/Nando75/SplashProbe/tree/main/flashingTool/
The folder contains the heltec tool for flashing the CubeCellflash.exe card.

The files **flashBoard_IT.bat** and **flashBoard_EN.bat** also facilitate uploading the firmware in a Windows environment.
Simply insert the CubeCell board and run the .bat file.
For other operating systems, download the appropriate tool from the official website:
https://resource.heltec.cn/download/
(Download the CubeCell Flash Tool suitable for your operating system).

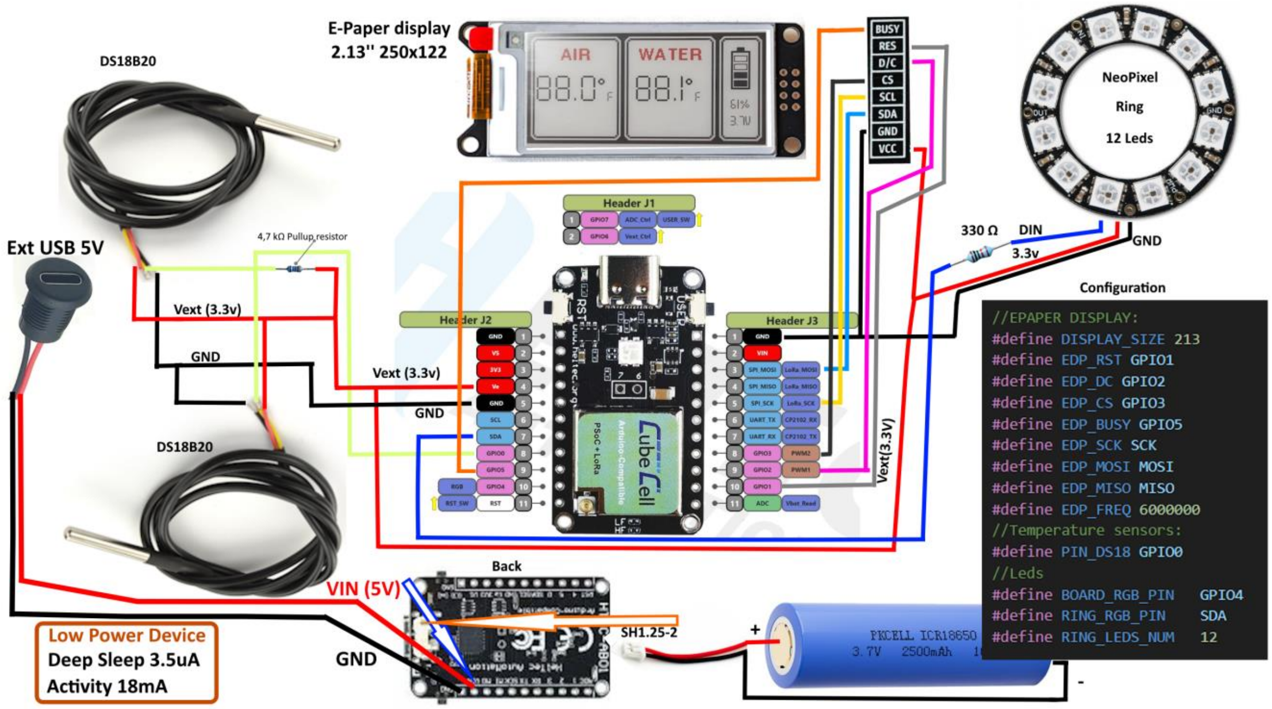
Firmware with **Celsius** degree and italian messages:
firmware_it.cyacd

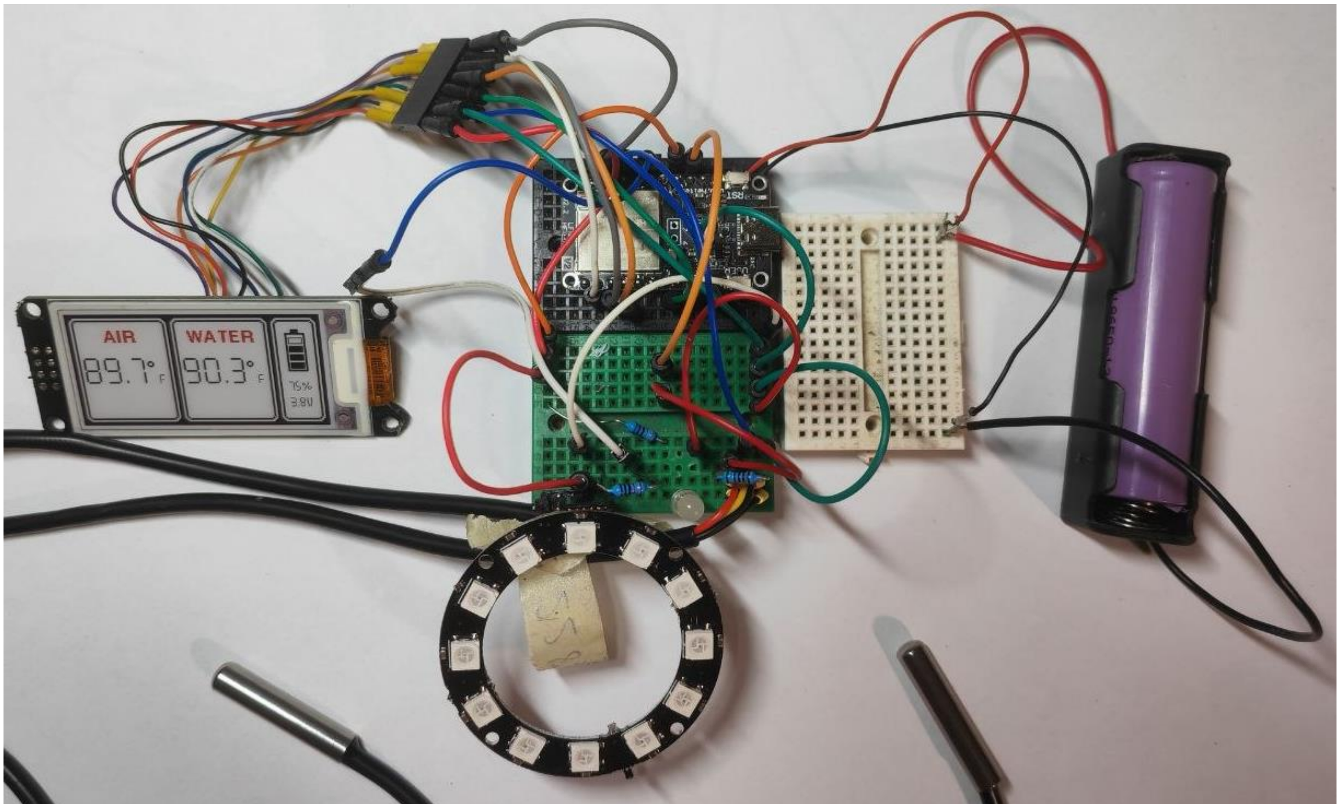
Firmware with F**ahrenheit** degree and english messages:
firmware_en.cyacd

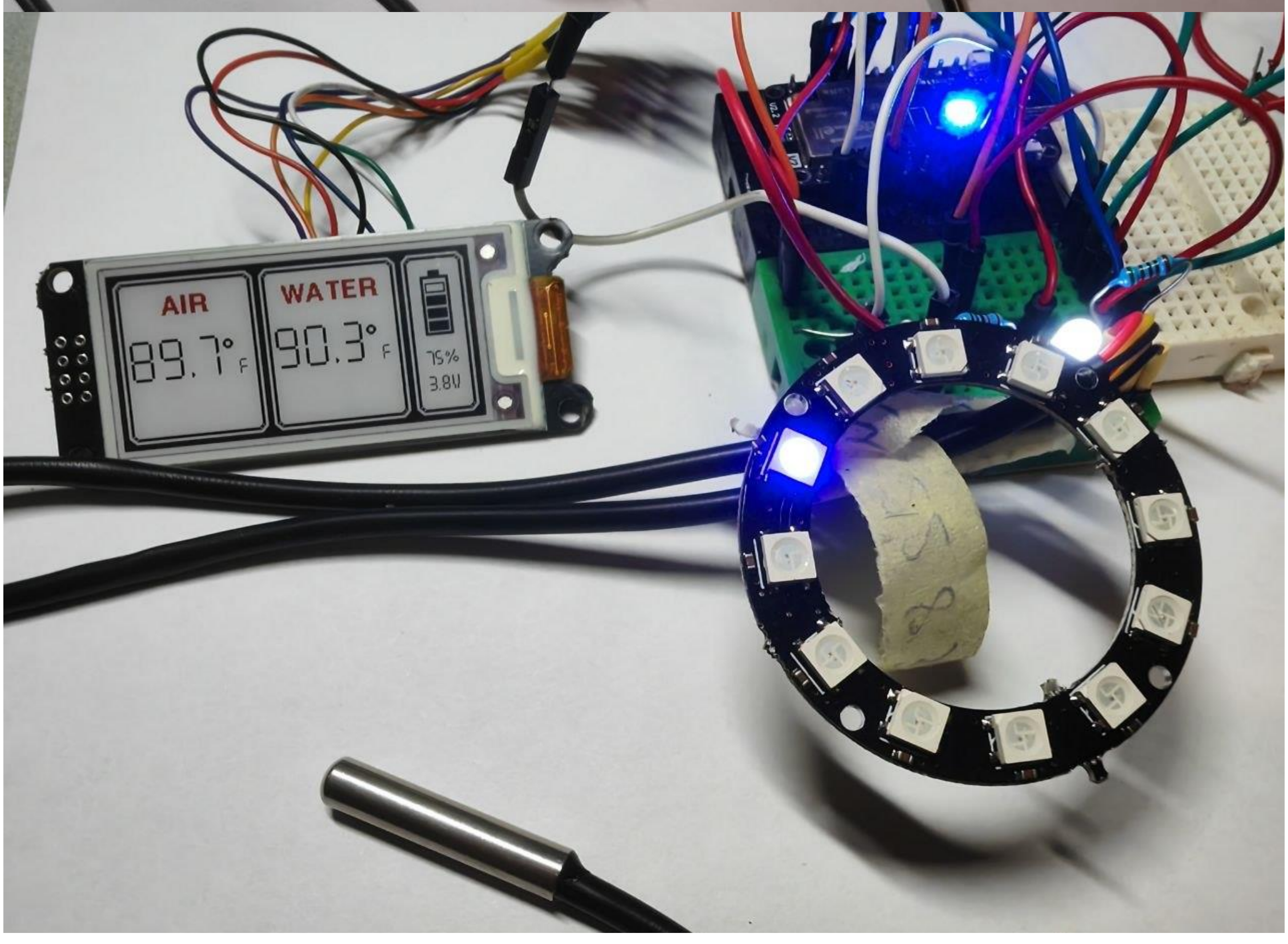
Otherwise, you can compile the sources with any customizations you want.
The project is developed with MS Visual Code and Platformio.

STEP 8: Circuit Wiring & Soldering

Wire and solder all electronic components (CubeCell board, 2.13" ePaper screen, 12-LED NeoPixel ring, 2x DS18B20 sensors, 330 Ω resistor, and 4.7 kΩ resistor) according to the provided circuit diagram.




AIR
WATER
89.7°F
90.3°F
75%
3.8V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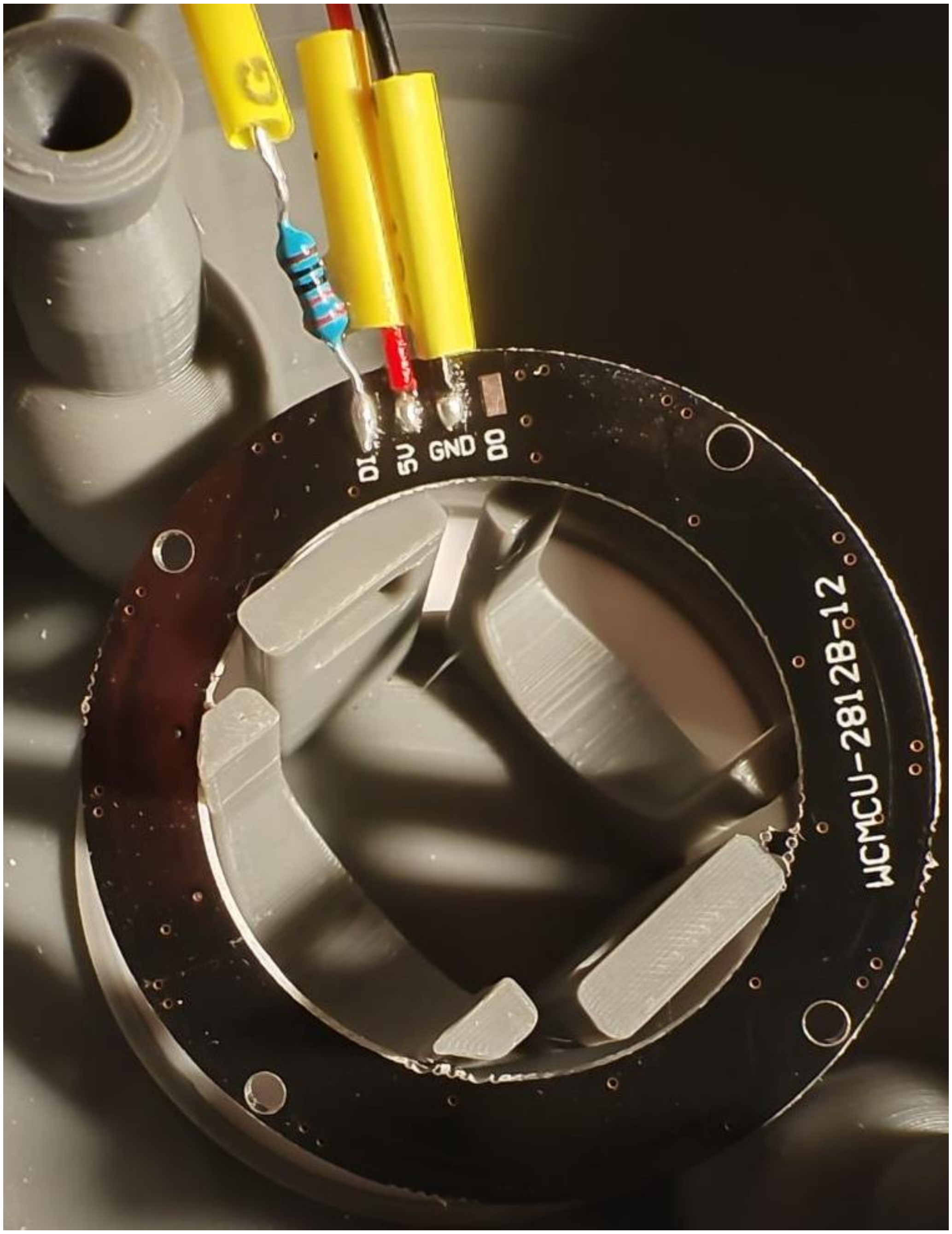
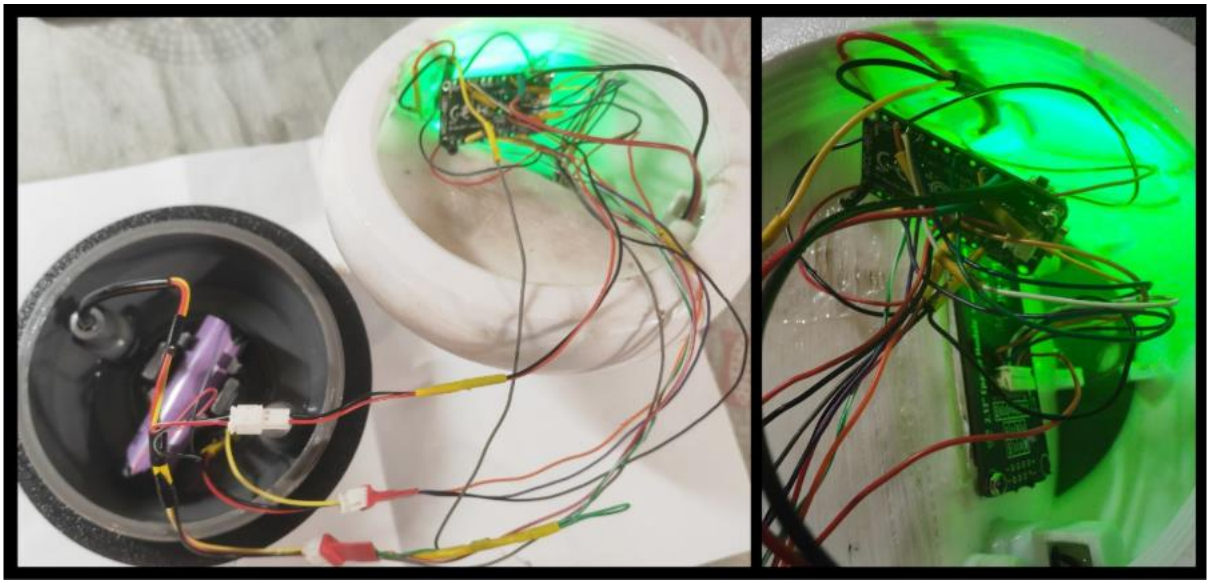


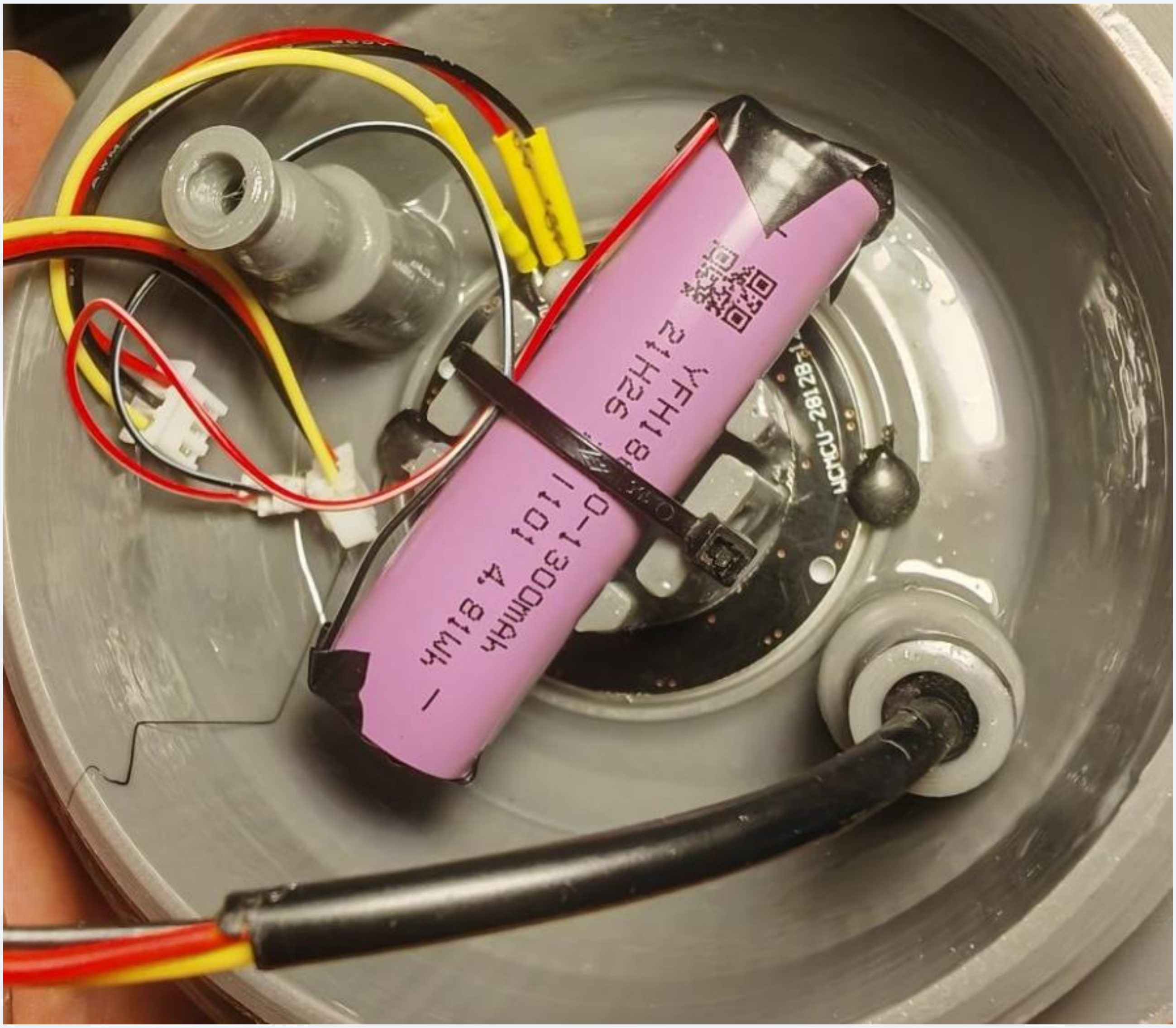
STEP 9: Display & Microcontroller Mounting

Mount the ePaper screen and the CubeCell board into their designated mounting slots inside the top dome. Secure them tightly using the provided tiny screws.



STEP 10: Battery Installation & Cable Management

1. Connect the 18650 LiPo battery to the board via its SH2 connector.
2. Seat the battery into the bottom housing of the base unit and lock it in place with a plastic zip-tie.
3. Tidy up all internal wiring using small zip-ties to prevent interference or pinching during closure.



I added a two-pin connector for the battery, a three-pin connector for the sensor, and one for the ring (optional). These connectors allow the two bodies to be separated by unscrewing them.

STEP 11: Final Enclosure & Waterproof Sealing

1. Insert the large TPU gasket onto the main threaded joint and screw the two main halves together tightly to prevent water ingress.
2. Install the small TPU gasket and USB cap onto the front panel, locking everything securely with two screws.

Congratulations! Your device is fully assembled, waterproofed, and ready for deployment!

Firmware Information

The board operates as a **State Machine**. It maintains a **Deep Sleep** state for ultra-low power consumption. Periodically (every 5 minutes), it wakes up via a timer to run the LED light sequence. Every 6 light-sequence cycles (every thirty minutes), it also reads the sensors and updates the e-Paper display. The main logical flow is easy to interpret and can be found in the **main.h** file.

All parameters and PIN configurations are defined in the **sys.h** file:

inline bool debug = true;
Prints debug information to the serial port; it can be set to false for release build.

inline Language Lang = en;
inline DataMode dataMode = Farfahrenheit;
Through parameters, the firmware can read and display temperatures in Celsius or Fahrenheit, with support for Italian (it), English (en), chinese (cn) and Spanish (es):

inline const uint32_t SLEEP_DURATION_MS = 5 * 30 * 1000UL; // 5 minutes
Light sequence using the 12-LED NeoPixel ring plus the board LED located under the alien graphic on the dome.

#define SENSORS_EVERY_N_CYCLES 6
Every 6 light-sequence cycles (30 minutes), it reads the sensors and updates the display.

Also in **sys.h**, there is the option to extend words used in different languages:
inline const char* airText[] = {"ARIA", "AIR", "AIRE"};
inline const char* waterText[] = {"ACQUA", "WATER", "AGUA"};
inline const char* lowBatt[] = {"Batteria Scarica", "Low Battery", "Batería baja"};

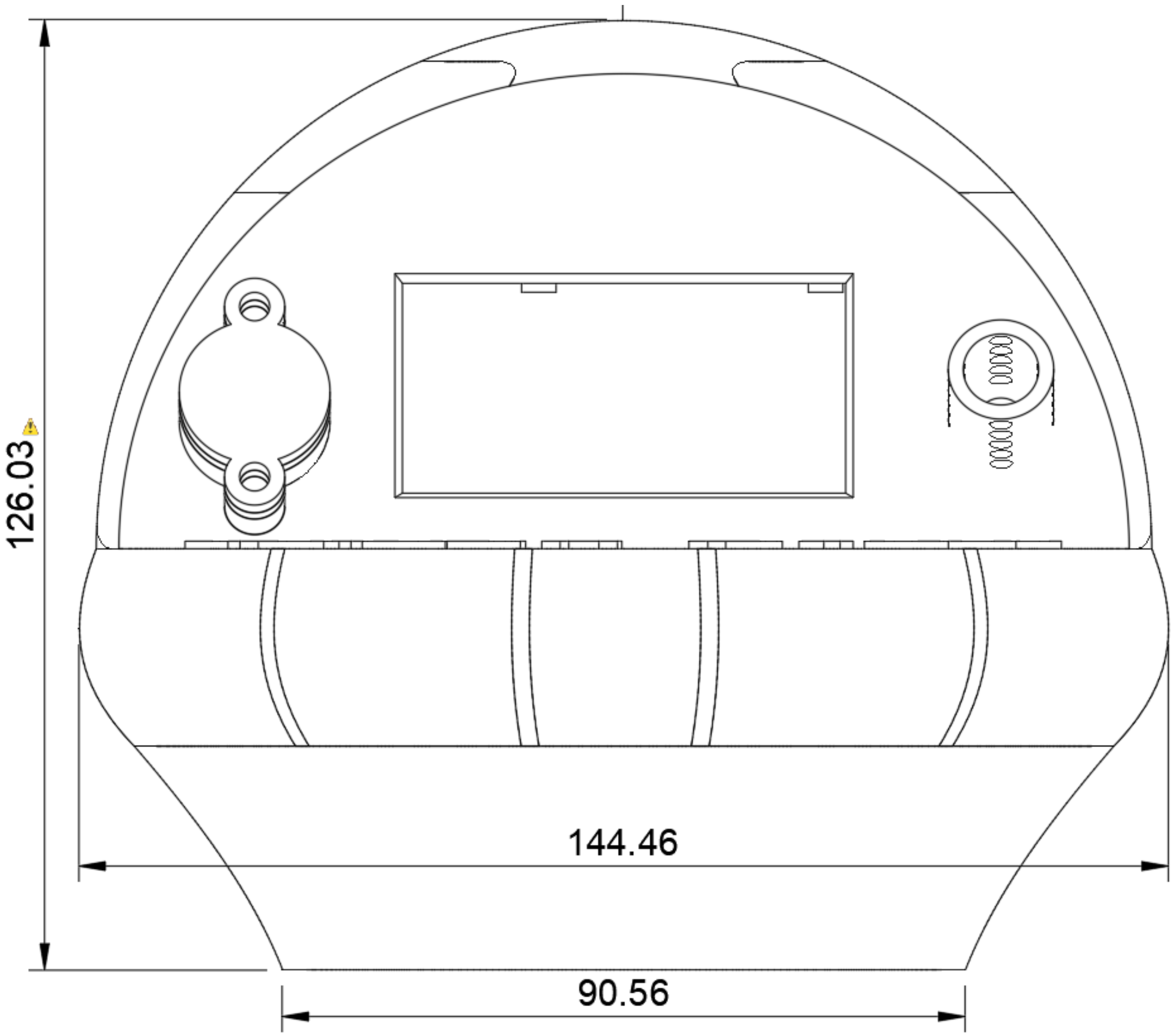
The two DS18B20 sensors return the actual temperature detected. In case a slight calibration of the values is needed, the **DS18Manager** class contains the following variables in its .h file:
float _airOffset = 0.0f;
float _waterOffset = 0.0f;
By adjusting these values either positively or negatively, it is possible to "fine-tune" the temperatures, perhaps by referencing another reliable and accurate thermometer.

All the code is organized into **Manager classes**. For instance, there is a DeepSleep Manager that handles the wake-up timer and enables/disables Vext, a Ring Manager for the lights, a Display Manager for the display, and so on.

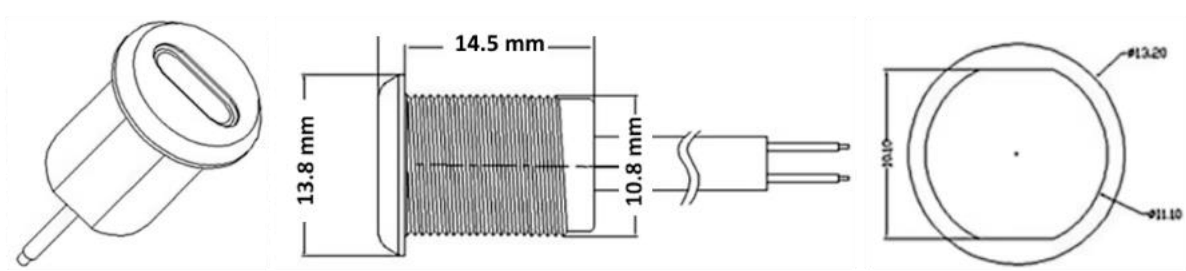
Components used:
Board Heltec Cubecell AB01 V2
Display: WeAct 2.13" E-Paper display (White/Black/Red)
External USB Port: 2-pin
ensors: 2 x DS18B20 temperature sensors
LED Ring: NeoPixel ring with 12 LEDs
Plexiglass: A3 sheet (297 x 420 mm), 1.8 - 2 mm thickness
Two-component epoxy resin (Resin Pro)
Glue for Plexiglass on PLA (Pattex Power Epoxy)

Additional materials:
1 x 18650 LiPo battery (1300 mAh)
1 x 330 Ω resistor
1 x 4.7 kΩ resistor
Wires, solder, and soldering iron.

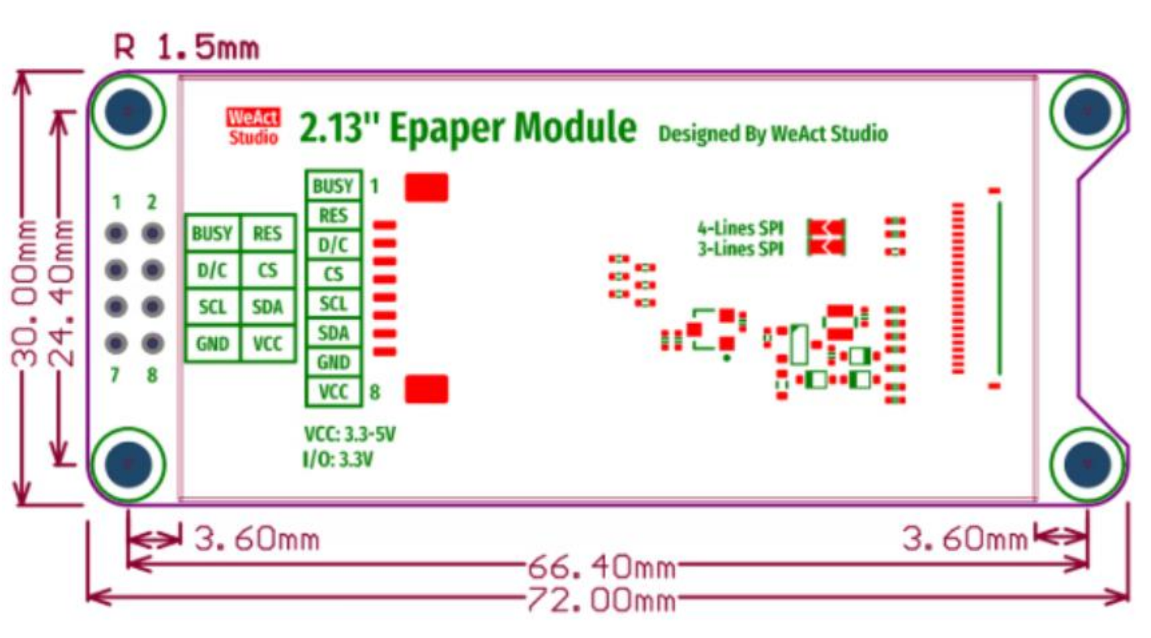
Measures and dimensions



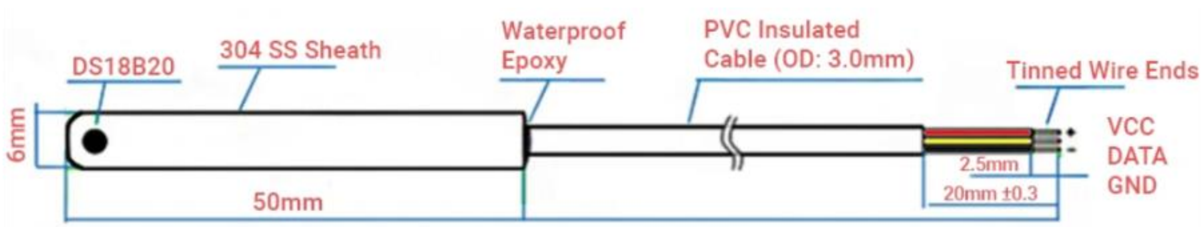
USB



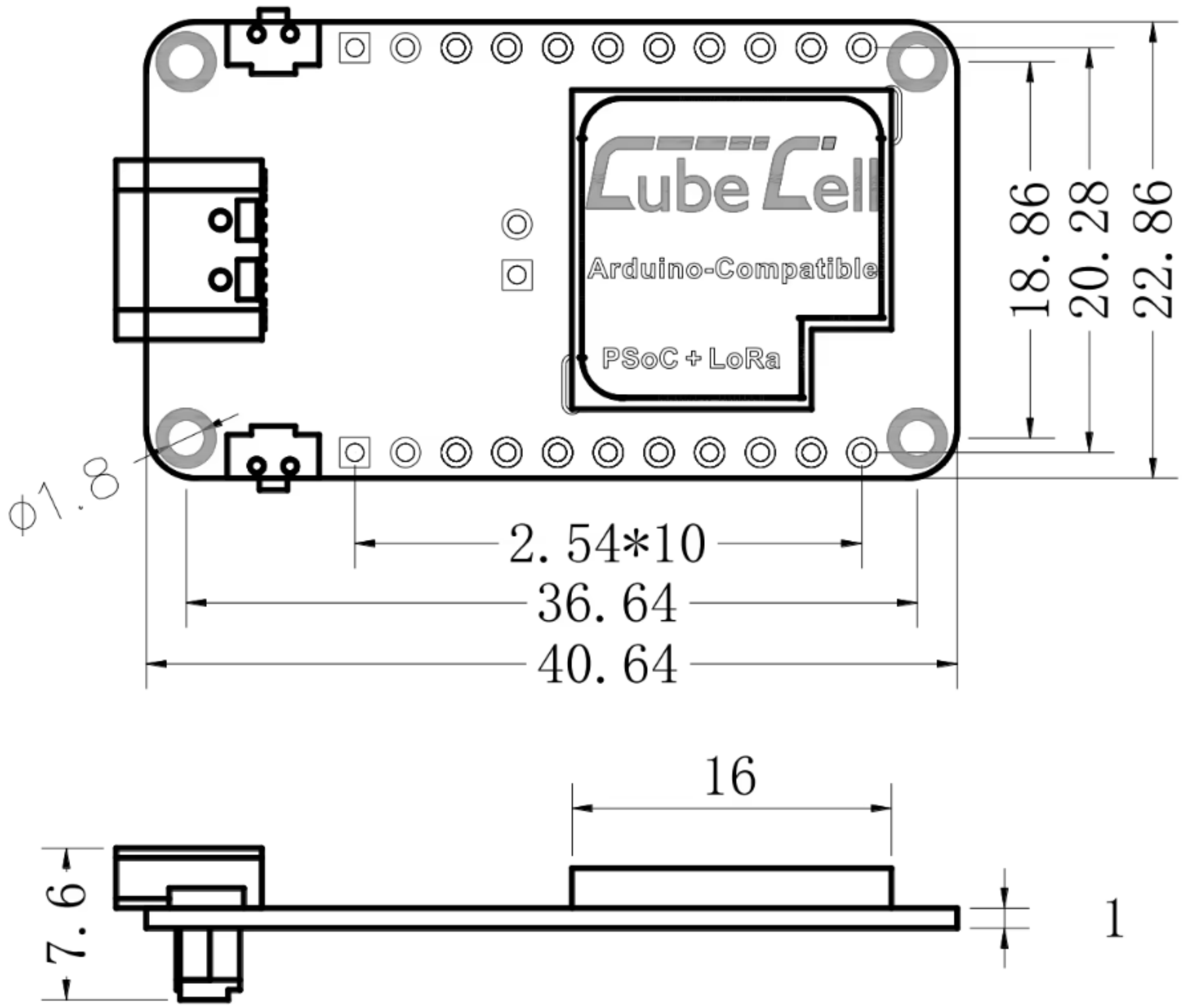
EPAPER



Sensore Temperatura DS18B20:



Cubecell AD01 V2:



Cube Cell
Arduino-Compatible
PSoC + LoRa
18.86
20.28
22.86
Φ1.8
2.54*10
36.64
40.64
16
7.6
1

18650 Battery Size